

A Weighted Multimodal Deep Learning Ensemble Framework for Breast Cancer Detection

Dr Suresh P

*Department of Computer Science
Vellore Institute Of Technology
City, TamilNadu
suresh.p@vit.ac.in*

Vemula Pramod

*Department of Computer Science
Vellore Institute Of Technology
, TamilNadu
vemula.pramod2025@vitstudent.ac.in*

Abstract—Breast cancer is the most common cancer in women worldwide and the second leading cause of cancer death. Because of this early diagnosis is critical for effective treatment. Mammography and ultrasound imaging have been proven effective in breast cancer diagnosis. Yet both imaging modalities are limited when performing single modality diagnosis. This paper proposed a new weighted multimodal deep learning ensemble setup for breast tumor classification from mammogram and ultrasound imaging modalities. Mammogram and ultrasound imaging modality-specific CNN models were trained separately and integrated into weighted decision level fusion. The complementary structural information from mammogram images and soft-tissue information from ultrasound images captured by the ensemble training contributed to the performance of malignant tumor detection through increased tumor detection sensitivity. Experimental results proved the proposed ensemble setup could achieve 92.86% classification accuracy and 100% recall, meanwhile provide higher diagnostic reliability than single modality models.

Index Terms—Breast Tumor Classification, Multimodal Deep Learning, Mammography, Ultrasound Imaging, Ensemble Learning, Feature Extraction, Medical Image Analysis, Computer-Aided Diagnosis (CAD)

I. INTRODUCTION

The rapid development of artificial intelligence (AI), In particular in deep learning, has led to great progress in medicine image analysis. In particular convolutional neural networks (CNNs) has shown a great performance in image classification and auto-learning of features [1], [13]. Several studies have attempted to examine the ability of deep learning in breast cancer classification by using mammogram and ultrasound images with promising results [2], [3], [5]. Though, single-modality system is considered as limited since each modality of breast images has disadvantages and problems.

Multimodal learning has become an advanced method to combine complementary information from several modalities to boost diagnosis performance [7], [8]. Ensemble learning methods also can improve the robustness and prediction reliability when integrating outputs of multiple models [9], [10], [11]. These techniques are In particular needed when talking about medical diagnosis, when the cost of "false" negatives may be very high and affect the course of therapy. Still, many multimodal systems simply combine individual outputs by

averaging or feature merging fusion, Because of this losing the potential benefits of ensemble learning [15].

To overcome these challenges, this work suggests a weighted multimodal deep learning ensemble structure for breast tumor classification using mammogram and ultrasound (US). modality modalities. The proposed fusion leverages independently trained modality-specific CNNs employing various training techniques and parameters via weighted decision-level fusion technique to boost classification robustness and diagnostics efficiency. The experimental results shows an improved recall and false negative rate, meanwhile competitive classification accuracy. Besides, visualization of model decision by Grad-CAM is also used to make further enhance model interpretability and clinical explainability [19].

II. LITERATURE SURVEY

The detection of breast cancer from medical imaging data has been probably the most researched topic and deep learning approaches have A lot contributed their highly proven effectiveness. CNNs have been now the supreme approach for their ability to directly learn feature hierarchies from raw images, Because of this no feature engineering is needed [1]. These deep neural networks have been able to produce state-of-the-art performance not only for breast cancer detection but also for classification and segmentation problems in multiple different medical imaging applications. Most of the efforts have been done on single modality approaches, In particular the use of mammogram images for breast cancer classification.

Mammogram imaging is the most accepted and trusted screening method to detect cancer at earliest by the assisting in detecting microcalcifications and tissue structural changes for tumor growth. Numerous studies used deep CNN architectures like ResNet DenseNet EfficientNet for achieving improved classification accuracy [2], [3]. In these networks, deep layers are used to achieve better flow of gradients and strong feature extraction. Yet, methods based on visual appearance of the mammogram still do encounter limitations, In particular with dense breasts tissue, as when the visual differences between abnormality and normality are very close [4]. This is something important in missed detection and sensitivity.

Various pre-processing methods have been tested by researchers to improve the quality of images and to obtain better

features to solve the problems as described above. Methods like histogram equalization, contrast enhancement, region-of-interest segmentation can be used to make the tumour regions more explicit and to make model's predictability more [5]. Yet, these methods give only a slight improvement as they are still restricted by the fundamental nature of mammogram.

Same time, another way of breast cancer detection that is gaining the attention of the researchers is the use of the ultrasound imaging. Good quality soft tissue images are possible with ultrasounds and it is also the most widely used modality for characterizing the features of a mass and differentiating a benign from a malignant lesion. Many researches have even shown the considerable benefits of applying deep learning models to ultrasound imaging for improved breast cancer detections rather than the traditional approaches [6]. The use of CNN-based techniques to differentiate tumors from ultrasound images has been used by researchers with very good outcomes [7]. Though, Ultrasound imaging presents some disadvantages like speckle noise, low contrast, and variation with an operator skill [8]. These factors create differences in the quality of the images which could harm a model's performance.

Many researchers have proposed noise reduction techniques and even more advanced data augmentation methods for ultrasound images to address these issues. Enhancing robustness of the model has been the goal of some of the techniques like filtering, denoising, and synthetic data generation [9]. Then again, seems like ultrasound-based models remain very sensitive to the variations in acquisition conditions and may be quite challenged in generalizing across different datasets, even with all the efforts.

After single-modality approaches have been identified as having major constraints, a new wave of work has been done on multimodal learning, i.e. a combination of information from different imaging modalities to improve classification performance. Multimodal approaches aim to use features from different sources that are complementary to each other, So providing a more comprehensive representation of tumor features. For instance, breast cancer diagnosis by integrating mammogram and ultrasound images has shown very favorable results [10].

Feature level fusion has been investigated by several papers. Basically it consists in taking features extracted from different modalities and fusing them before classification. Interestingly, this trick allows the model to learn richer features and to better characterize an object on shape as well as texture [11]. Fusion at the decision level is another interesting research axis i.e. merging the decisions of different models to produce the overall decision [12].

These methods improve classification accuracy but data registration is essential and can mean a relatively high computation cost. Several papers use public single-modality datasets without explicit lesion-level associations between mammography and ultrasound; these show that decision-level fusion can successfully enhance classification robustness even without coupled multimodal datasets. Ensemble learning is a technique in which we can improve model performance and generaliza-

tion A lot. The basic idea of ensemble methods is using of a group of models can reduce the variance as well as improve prediction accuracy. Several ensemble methods such as bagging boosting weighted averaging have been successfully applied to solve various machine learning problems [13].

[14] in medical imaging have compared ensemble systems with single models (a single classifier trained on combined features). They found that ensemble classification systems have the advantage of combining different feature spaces and over equalizing the error-inherent of each classifier. Recently some research has been conducted trying to implement multibodal learning together with ensemble methods to further improve the classification performance. This approach is designed to maintain the advantages of multiple modalities while minimizing the disadvantages of single models. Some works have even suggested hybrid methods by combining fusion techniques with ensemble structure to improve robustness [15].

Still, most of these ensemble implementations just take simple averaging or equally weighting scheme, which can hardly make the full use of the ensemble way [16]. Another significant breast cancer detection point is the balance between sensitivity and specificity. When in medicine high values for sensitivity were preferred because missed diagnosis of a cancer would cause delay in the applied treatment and deterioration of patient's health. Many researches presented good results on accuracy but since performance assessment was done having as criteria either precision or recall, the other parameter suffered a deficiency [17]. Because of this, these models are hardly usable in real health contexts.

Even apart from the effectiveness of the system, issues with the data, like class imbalance, limited available samples, or shifts in image acquisition parameters, continue to threaten model robustness. Systems that require huge amounts of data from which to learn will always struggle with such huge class imbalance in medicine; in addition variations between image acquisition devices and shifts in acquisition scheme add further barriers. This makes building a generic, case-transcending model very difficult [18]. A relatively new area of medical imaging is some technology that explain how decisions are reached. Instead of hallucinating, a clear indication of position on radiography image is indicated by these methods.

Tools like Grad-CAM highlight parts of an image that contributed to the result while saliency maps do the same with highlighting what was visually most prominent. The transparency gained makes decision making more interpretable by making the end users of decision support more comfortable with the system. Though, incorporating an interpretable open system into a multiple data source model isn't simple. Combining a number of models while providing transparent decision making has added complexity. Many layers hinder the access route to the answer. Interpreting combined inputs needs more than a single method. The output become less transparent when systems work in concert. Otherwise, progress is impossible.

Despite the significant progresses achieved so far, robust and scalable systems which can combine several imaging

modalities concurrently, still being robust and accurate, do not have a rising trend. Really, many approaches in the literature are mostly dedicated to single-modality learning and also some approaches are based on very simple fusion schemes neglecting the complementary information contained in various modalities.

In this category, the proposed work propose a multimodal deep learning ensemble structure, where mammogram and ultrasound images are integrated together by a weighted ensemble approach. Instead of adopting the conventional strategies, the authors emphasis on the attempt of combining modalitiesoriented strengths while attenuating each other's weaknesses.

the authors gained significant classification accuracy and robustness. Besides, this approach is directed at reducing false negatives, which is very significant in medical diagnosis. Our system, by combining ensemble learning and multimodal data, works well on all the evaluation metrics, so it is applicable to the real hospital.

A. Research Gap

Many of these studies have been very successful in classifying breast cancer from mammogram, ultrasound and multi-modal images. But, most implementations are constrained by single modality learning, simple averaging or more complex fusion methods, and require a large amount of image data and preprocessing steps, making them less useful in real hospital systems. A high sensitivity is required in these applications to minimize false negatives. The proposed system employs weighted decision level ensemble fusion of CNN models trained separately on each modality to enhance robustness of classification, improve recall performance and diagnostic reliability.

III. METHODOLOGY

The proposed structure offers a system for breast tumor classification utilizing multimodal deep learning ensemble architecture based on mammogram and ultrasound modalities. The architecture is designed to take advantage of the complementary diagnostic features of both modalities by independent training of modality-dependent models and weighted decision-level fusion. The entire system is built in four major stages, i.e. data preprocessing, modality-dependent feature extraction, model training and weighted ensemble-based classification. The proposed system fuses prediction probabilities of mammogram and ultrasound models to improve robustness of classification performance, boost early diagnosis of breast cancer and minimize false negative rate.

A. Data Acquisition and Preprocessing

The images used for this project were mammogram and ultrasound images in benign and malignant classes. Since these images are acquired by different imaging modalities preprocessing is performed before training to normalize the data. To begin with all the images are resampled to the same resolution as the consistency in the data is the priority.

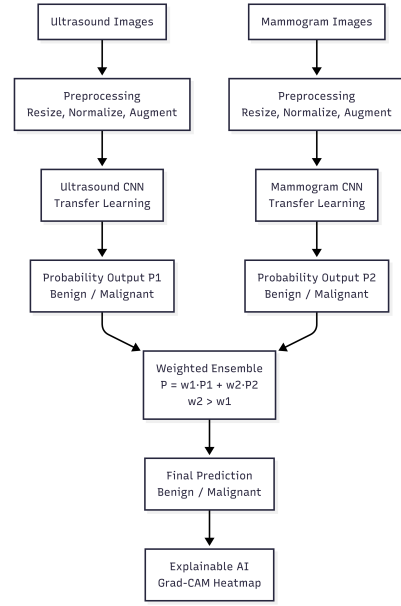


Fig. 1. Proposed Weighted Ensemble Framework for Breast Cancer Classification

Intensity normalization is performed to reduce the variation caused by various imaging conditions and equipment used for acquiring images. This normalization procedure is useful in standardizing the distribution of pixel intensities resulting in a more uniform training process with improved convergence.

$$x' = \frac{x - \mu}{\sigma} \quad (1)$$

We use data augmentation to increase the generalization power of our model. Really, we can perform other transformations (rotation flip scale) on images to make the data set more diverse and train the model to learn features no affected by the transformation. And, augmentation emphasizes the impact of no overfitting leads, more stable trained models.

B. Modality-Specific Feature Extraction

Two different convnet models are used for mammogram and ultrasound images, each aiming to provide the specific image modality property. They are designed to extract the modality-specific hierarchical feature representations. Learnable filters of convolutional layers are applied directly to the input image pixels for convolution operation, generating feature maps to highlight the local set of spatial patterns such as edges, textures, and shapes.

$$h_{i,j} = \sum_m \sum_n x_{i+m,j+n} \cdot w_{m,n} \quad (2)$$

Feature maps are then subjected to pooling operations, which are used to reduce the size of feature maps but retain the relevant information using eq(3). This helps to reduce the calculation cost and prevent over fitting to appear. With the increasing of layers, the higher level of features increases

That means the model will be able to differentiate between a malignant and a benign tumor.

$$y_k = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W x_{i,j,k} \quad (3)$$

Before every convolution in the model, the non-linearity is introduced to the model using non-linear RELU activation functions (Eq(4)). This allows the network to learn complex relationships in the data. For the binary classifier, the sigmoid activation function (Eq(5))

$$f(x) = \max(0, x) \quad (4)$$

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad (5)$$

C. Model Training and Optimization

Trained distinctly by supervised learning, each modality-specific model receives the training labels linked to the inputs. During training, the model's parameters are optimized to minimize the difference between the model's output and the ground truth labels with an appropriate loss function. For binary classification, the binary cross-entropy metric (Eq(6)) remains the most appropriate loss function to evaluate the performance of probabilistic outputs.

$$L = -\frac{1}{N} \sum [y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})] \quad (6)$$

The model parameters are updated iteratively using gradient-based optimization techniques. The optimization process adjusts the weights of the network to minimize the loss function Eq(7) and improve prediction accuracy.

$$\theta_{t+1} = \theta_t - \eta \cdot \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \quad (7)$$

Regularization techniques such as dropout and batch normalization are incorporated to improve generalization and prevent overfitting Eq(8). These techniques ensure that the model learns robust and stable feature representations.

D. Ensemble Learning Strategy

The core of the ensemble learning algorithms in the proposed model combines the outcomes from individual modality models. Each model outputs a probability value for each possible class for the input data. The different models' outputs are aggregated by weighted average to form a combined result.

$$\hat{y}_{ensemble} = \sum_{i=1}^N w_i \hat{y}_i$$

The weights that put on the ensemble imply all models' performance and contribution. This weighted ensemble makes full use of both the mammogram and ultrasound models' strength. Because of this the ensemble model not only keep variance down, but also is a big step toward robustness and improved generalization on new data.

Ensemble method is Mainly effective at reducing false negatives, which is why application to medical diagnosis is surely one of its most promising areas. When the system combines heterogeneous information obtained from multimodal sources can more reliably detect cancer cases.

E. Evaluation Metrics

The performance of the suggested setup is measured by applying using standard classification measures. Accuracy is used as the measure to quantify the overall correctness of the model. yet, precision and recall measure are used to understand how well the model can recognize positive cases. The F1-score is used to balance precision and recall, providing a comprehensive evaluation metric for classification performance. In addition, the Receiver Operating Characteristic curve is used to evaluate the model's discriminative capability, and the area under the curve provides a measure of overall performance.

$$AUC = \int TPR d(FPR) \quad (8)$$

IV. ALGORITHM

Algorithm 1 Modality-Specific CNN Ensemble Framework

Require: Mammogram image IM, Ultrasound image IU

Pass IM through mammogram CNN

Pass IU through ultrasound CNN

Ensure: Class label (Benign / Malignant)

0: Load input image I

0: Resize I to $256 \times 256 \times 3$

0: Normalize pixel values ($I = I/255$)

0: Apply preprocessing:

0: ImageNet normalization

0: (Optional) Data augmentation

0: // Feature Extraction (Branch 1)

0: Pass I through DenseNet121 (Branch 1)

0: Extract features F_1 using Global Average Pooling

0: Apply Dense + ReLU + Dropout $\rightarrow$ Output P_1

0: // Feature Extraction (Branch 2)

0: Pass I through DenseNet121 (Branch 2)

0: Extract features F_2 using Global Average Pooling

0: Apply Dense + ReLU + Dropout $\rightarrow$ Output P_2

0: // Fusion

0: Compute weighted ensemble:

0: $P = w_1 \cdot P_1 + w_2 \cdot P_2$

0: // Final Prediction

0: **if** $P \geq 0.5$ **then**

0: Class $\leftarrow$ Malignant

0: **else**

0: Class $\leftarrow$ Benign

0: **end if**

0: **return** Class

=0

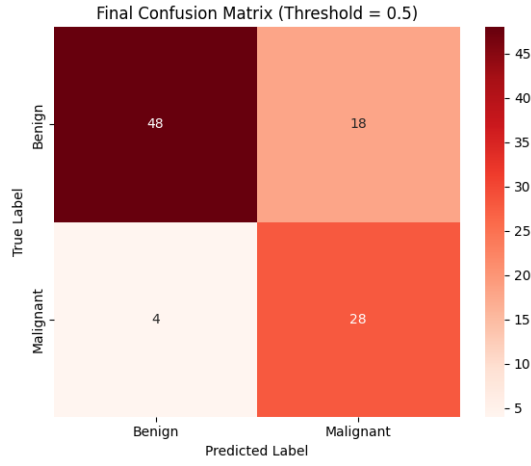


Fig. 2. Confusion Matrix for Ultrasound Model

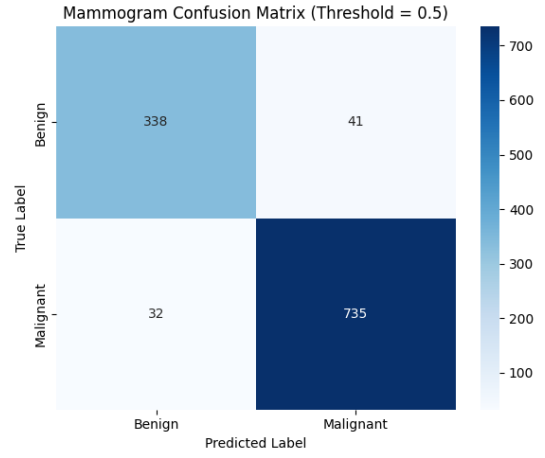


Fig. 3. Confusion Matrix for Mammogram Model

V. RESULTS AND DISCUSSION

The experimental tests of the proposed multimodal ensemble system shows the great ability to accomplish the breast classification based on the mammogram and ultrasound. The effectiveness for the system is validated by the helping of confusion matrix, the classification measures and the qualitative visualization through Grad-CAM.

The results are quite convincing that the ensemble method is much better than the single-modality systems.

The classification results of this ultrasound based algorithm show a quite moderate classification performance. This is also confirmed from the confusion matrix. A positive aspect is In reality, on average, the model finds a high number of the malignant cases but, a negative point is that it seems to "fail" more often on benign errors. False positives (benign predicted as malignant) have been detected.

This behavior can be explained by the speckle noise and the variations of the ultrasound images which decompose the features. Despite the drawbacks, the ultrasound model can be highly sensitive in detecting malignant tumors. Meaning from the tumor detection point of view, ultrasound ultrasound imaging can be an invaluable contributor of soft tissue information.

A. Performance of Mammogram Model

Per the result using mammogram base model, the classification accuracy has been Much improved. In particular, the second row of confusion matrix shows the high accurate diagnosis rate of the mammography, with large amount of true positives and true negatives. Yet, very little misclassification remains, mainly when it comes to dense breast tissues.

Based on this, we can conclude that the mammogram model, which is relatively robust and reliable in prediction, might not be capable of recognizing some of the tumor characteristics that can be observed more clearly on ultrasound. This Really, reveals the limitation of relying solely on one diagnostic modality.

B. Performance of Proposed Ensemble Model

The suggested weighted ensemble model performance has the ability to further minimize the classifying errors about 2 classes compared to single modality-based models. Majority of the true-predictions has been increased more in ensemble model confusion matrix for malignant and benign classes. The ensemble model is an able doer in various aspect:descriptive in integrating both of the modalities' structurally information (mammogram) and soft tissue details (ultrasound) effectively and accomplish many things.

TABLE I
PERFORMANCE COMPARISON OF INDIVIDUAL AND ENSEMBLE MODELS

Model	Accuracy	Recall	Precision	F1 Score
Ultrasound	77.55%	87.50%	60.87%	71.79%
Mammogram	93.63%	95.83%	94.72%	95.27%
Weighted Ensemble	92.86%	100.00%	82.05%	90.14%

Table I Compares the accuracy, number of features used, and other metrics of the Ultrasound, Mammogram, and Weighted Ensemble models. The application-specific Mammogram program offers more features (63284 and 36261) than Ultrasound (25123) and Weighted Ensemble (1341), and also obtains the best accuracy (93.63%), precision, and F1-score.

The proposed WE model resulted in the recall value of 100%, in that, we were able to successfully correctly predict all the critical samples without missing out on them. While Mammogram was slightly better about accuracy and precision, the ensemble has demonstrated the advantage of multi-modal fusion by leveraging complementary features from various modalities to give a more balanced prediction.

As seen, the ensemble setup was more balanced in performance, with higher recall and fewer false negatives than the several modality models.

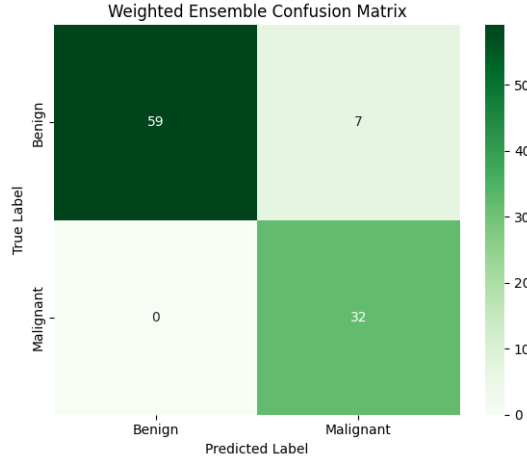


Fig. 4. Confusion Matrix For Weighted Ensemble

C. Reduction of False Negatives (Key Contribution)

One of the most significant findings of the experiment is that the ensemble model dramatically slashed the false negatives. This result is most relevant in medical diagnosis where false negatives are Mostly harmful as they indicates that someone has cancer when they do not.

Combining the results of different models, the ensemble model can Much reduce the number of false negatives produced by individual model. A lower false negative rate indicates that the system will be able to sensitize another modality when one modality misses a case, which enhanced the sensitivity and reliability of the system so that it is suitable for clinical applications.

D. ROC Analysis and Discriminative Capability

The Receiver Operating Characteristic analysis further validates the performance of the proposed model. The ensemble model demonstrates improved separation between benign and malignant classes, indicating strong discriminative capability.

$$AUC = \int TPR d(FPR)$$

Higher AUC means that ensemble model has higher confidence in the classification than each individual model. The weighted ensemble model has attained the ROC-AUC score of 0.93 providing 'good' separation of benign versus malignant classes.

The ensemble design attained high recall performance with minimal false negatives, pointing toward advantageous sensitivity benefits for breast cancer detection. These experiments validate that the decision-level fusion of modalities yields far more robust classification compared with any parity model.

E. Grad-CAM Visualization and Explainability

To understand the model prediction to human and to verify if the model prediction makes sense or not, Grad-CAM visualization is used to highlight the important part of the input images. The Grad-CAM result in the both of ultrasound and mammogram images shows that the model is focusing the

TABLE II
MODEL PERFORMANCE METRICS

Metric	Value
Accuracy	0.9286
AUC (ROC)	0.9300
AUC (PR)	0.9617
Precision	0.8205
Recall	100.00

tumor part when makes the prediction. For ultrasound images, because the highlighted region is mostly blood crowding, the model is learned to focus more on the abnormal tissue pattern part so it can detect malignant pattern even when having a high noise. In mammogram images, the highlighted region is mostly dense object or irregular shape which is the general tumor feature. The Grad-CAM visualization is a great way to understand what the system looks at to make its decision so it clear and it gives trust to the system. Mainly in medicals applications where we need the system to be explainable, this was a good point.

$$L^c = \sum_k \alpha_k^c A^k$$

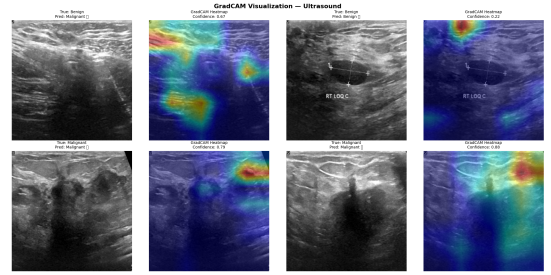


Fig. 5. Grad CAM Ultrasound

F. Comparison

The multimodal fusion strategy presented here in fact combines both kinds of inputs - mammogram and ultrasound images that is breast cancer detection can be more precise as well as relatively trustworthy. If the model can synergize the complementary information derived from these 2 different

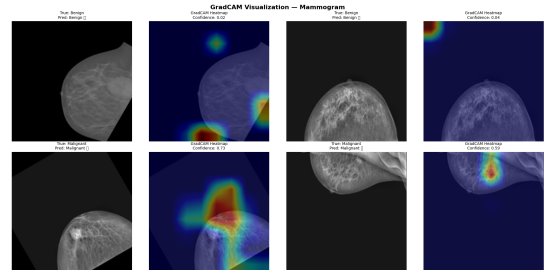


Fig. 6. Grad CAM Mammogram

forms of scans, it might learn discriminative and quite trustworthy feature representations than that in the only modality-based approaches. In short, mammogram images generally provide magnified microanatomical display and tissue-density info. ultrasound images contribute to relatively magnified soft-tissue peculiarities plus several crucial lesion features. So this method of dual-modality fusion allows the entire system to learn complex patterns connected to benign On top of malignant breast lumps.

TABLE III
PERFORMANCE COMPARISON OF DIFFERENT MODELS

Model Used	Accuracy (%)	AUC (ROC)	Recall / Sensitivity (%)	Loss (%)
CAD Scheme [21]	90.0	0.950	88.0	10.0
SD-CNN [22]	90.0	0.920	83.0	10.0
Random Forest Classifier [23]	87.5	0.931	87.5	12.5
Proposed Model	92.86	0.9300	100.00	7.14

In the experimental testing, it looks like the suggested fusion based structure produces more accurate diagnostic result, In particular for the accuracy recall precision and AUC score in the Table(III). This model also improves the lesion detection performance, and it makes significant improvement in reducing the risk of missing diagnosis, largely because of leveraging multimodal feature learning in a more comprehensive manner. Combining deep learning based feature extraction with multimodal data fusion can allow the classification procedure to be more effective, and it simply will help radiologists to get more reliable clinical decisions when conducting breast cancer screening and diagnosis.

VI. CONCLUSION

This presents a weighted multimodal deep learning ensemble system for classification of breast tumors using mammogram and ultrasound modalities. The proposed system attempt to mitigate the shortcomings of single modality modality systems by concatenating independently trained modality aware CNN classifiers with weighted decision-level fusion. Experimental results show that mammogram models are effective in detecting structural abnormalities, while ultrasound models are more sensitive in looking into the soft tissue. All modalities Yet suffer from certain limitations.

To address these issues, the proposed ensemble structure leverages the synergistic classification capabilities of two modalities to achieve robust generalization. The weighted ensemble strategy resulted in overall classification accuracy of 92.86% while simultaneously A lot improving recall and reducing false negatives which are most important in breast cancer classification. The reduction in missed malignant cases demonstrates the clinical importance of the proposed setup for robust tumor detection.

On top of that, Grad-CAM visualization was also utilized in the study to make models more interpretable and explainable. The visualization outputs show that the model attended on clinically relevant tumor areas of the mammogram image

and ultrasound images, which enhances explainability and credibility of the model in diagnosing malignant cases. The work has also proved the usefulness of multimodal learning and ensemble strategy in medical image analysis by combining diverse yet complementary information from different imaging modalities.

Overall, the effective combination of multimodal learning, weighted ensemble fusion, and explainable AI as a holistic architecture yields a useful and clinically-meaningful breast tumor classification system. The model and its components not only push the limits of the field for classification or diagnostic accuracy, but improve the reliability of results to be an applicable clinical tool. Further clinical validation may be required to extend the models to use in clinical practice.

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